

Lab 2 – Cosmic Crush Report

1. My Contributions

For this project, I developed a 3D game based on the Cosmic Crush concept using Unity and C#. I created the main game arena, including the floor and boundary walls, and the physics so that the spheres can bounce off the walls. I created the player sphere and implemented WASD movement using Unity's Input System. I also created an NPC sphere with random movement and converted it into a prefab so that 15 NPCs can be spawned automatically at random positions around the arena.

I implemented the main gameplay mechanic where spheres can collide with each other. When a larger sphere collides with a smaller sphere, the smaller sphere is removed and the larger sphere increases in size. I also implemented player mass loss while moving, meaning the player becomes slightly smaller when using the movement controls. A scoring system was added so that the player receives points for absorbing NPCs. I also added different colors based on sphere size and made the player blue so that it can be easily identified from the NPCs.

2. New Features and Techniques Learned

During this lab I learned more about Unity's Rigidbody physics, Sphere Colliders and Physics Materials. I learned how to use prefabs and instantiate multiple objects through C# scripts. I also gained experience using collision detection with `OnCollisionEnter()` and changing an object's scale during gameplay. Another new area I worked with was Unity's UI system to display the player's score and game-over messages. I also improved my understanding of Git and GitHub by committing and pushing the project while developing it.

3. Current Issues and Bugs

The main issue during development was getting the scoring and game-over systems working correctly. I also had an issue where NPCs could spawn too close to the player and cause the game to end immediately. I fixed this by preventing NPCs from spawning near the center of the arena. The game is currently functional, although some areas could be

improved with additional visual effects, better balancing of sphere sizes, and more polished UI.